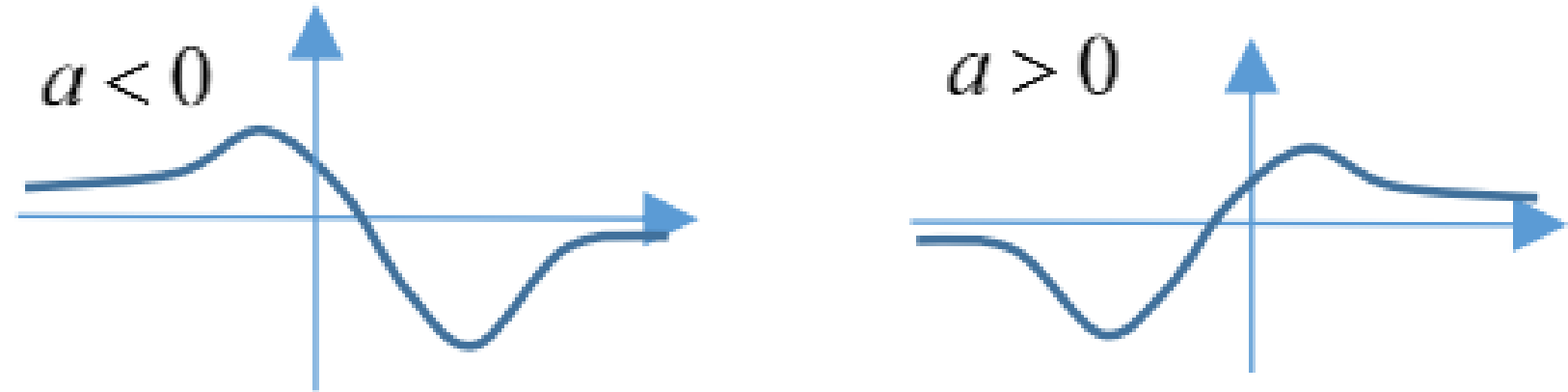
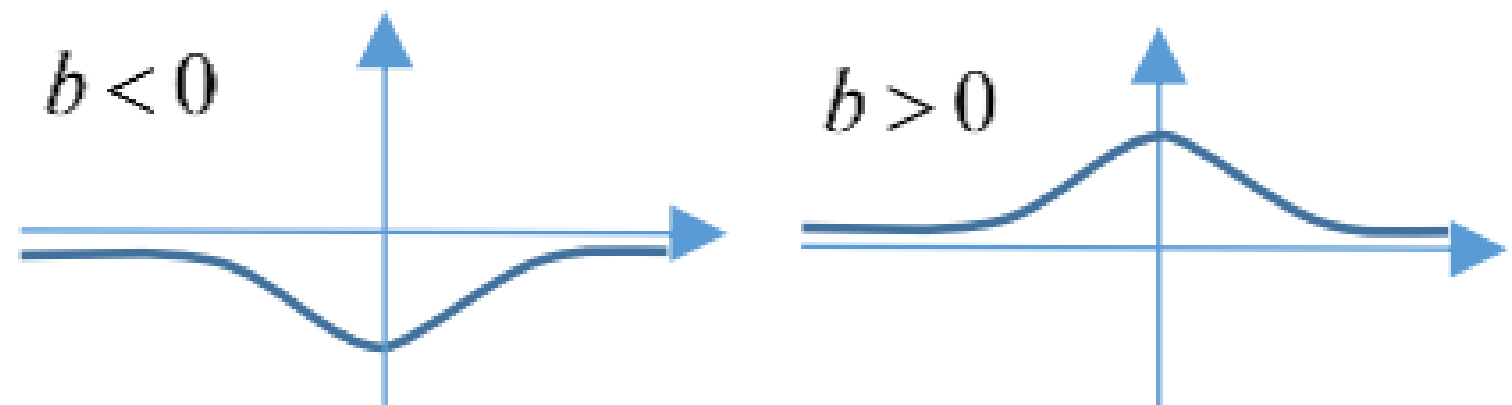
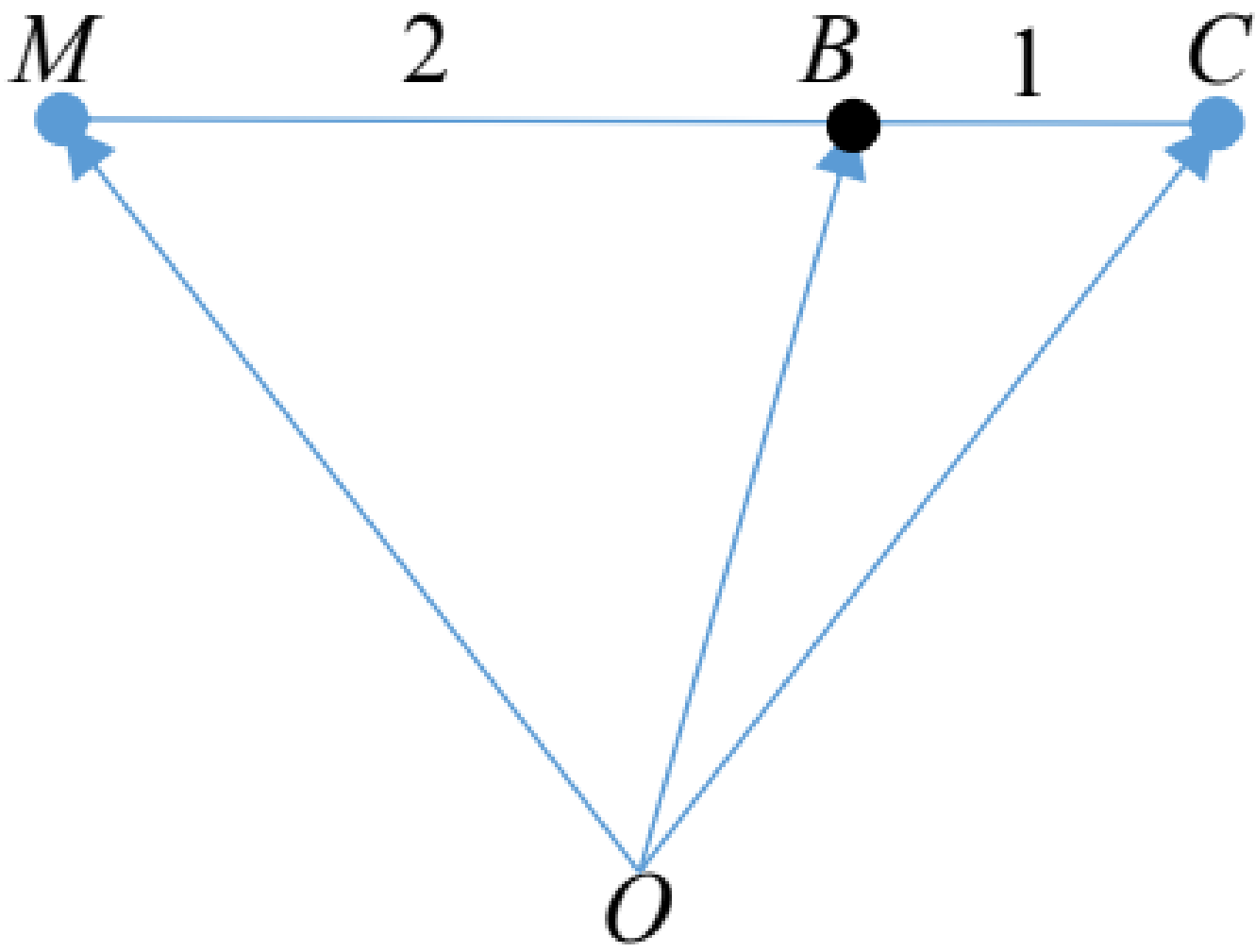


Qn	Solution	Comments
1	Let e, g, w be amount of electricity and gas in kWh and water in m^3 used by Mr Lim in June respectively. $0.1775e + 0.1715g + 2.852w = 143.06 - 36$ $0.1775e + 0.1715g + 2.852w = 107.06 \quad \text{---- (1)}$ $0.2365e + 0.2279g + 3.672w = 140.78 \quad \text{---- (2)}$ $0.1762e + 0.1698g + 2.741w = 144.96 - 40$ $0.1762e + 0.1698g + 2.741w = 104.96 \quad \text{---- (3)}$ By G.C., $e = 266.4558551 = 266$ $g = 115.0629925 = 115$ $w = 14.03603875 = 14.0$ Mr Lim used 266kWh of electricity, 115kWh of gas and 14.0 m^3 of water in June.	Students should define their variables before formulating their equations. The units (either dollars or cents) should be consistent for all three equations, else marks will not be awarded. Students should avoid rounding off errors such as: - Writing 14.0 (3s.f.) as just 14 - Double rounding off of 266.46 to 267 Students are reminded to read instructions on the cover page, which clearly stated that answers should be left in 3s.f. unless otherwise stated in the question.
2(i)	$\frac{d}{dx} \tan^{-1}(x^k) = \frac{kx^{k-1}}{1+x^{2k}}$	Students should not that in this case x is the variable and k is a constant. $\therefore \frac{d}{dx} x^k = kx^{k-1}$ Applying chain rule will result in $\frac{d}{dx} \tan^{-1}[f(x)] = \frac{f'(x)}{1+[f(x)]^2}$
2(ii)	$u = \tan^{-1}(x^k) \quad v' = x^{k-1}$ $u' = \frac{kx^{k-1}}{1+x^{2k}} \quad v = \frac{x^k}{k}$ $\int x^{k-1} \tan^{-1}(x^k) dx$ $= \frac{x^k}{k} \tan^{-1}(x^k) - \int \frac{x^{2k-1}}{1+x^{2k}} dx$ $= \frac{x^k}{k} \tan^{-1}(x^k) - \frac{1}{2k} \int \frac{2kx^{2k-1}}{1+x^{2k}} dx$ $= \frac{x^k}{k} \tan^{-1}(x^k) - \frac{1}{2k} \ln(1+x^{2k}) + c$ since $1+x^{2k} > 0$	Question is testing on use of Integration by Parts. Part (i) is a hint for students to take $u = \tan^{-1}(x^k)$ Again note that k is the constant and apply integration on $v' = x^{k-1}$ to obtain $v = \frac{x^k}{k}$. Students should make attempt to write out the expression for $uv - \int vu' dx$ and simplify the expression for $\int vu' dx$ as far as possible.

Qn	Solution	Comments
		Thereafter $\int \frac{x^{2k-1}}{1+x^{2k}} dx$ can be evaluated using the standard formula $\int \frac{f'(x)}{f(x)} dx$ by observing $f(x) = 1 + x^{2k}$ $f'(x) = 2kx^{2k-1}$
3(i)	$k = \frac{ax+b}{x^2+1}$ $kx^2 + k = ax + b$ $kx^2 - ax + (k-b) = 0 \quad \text{-----(1) (shown)}$	
3(ii)	For (1) to have real roots, $(-a)^2 - 4k(k-b) \geq 0$ $a^2 - 4k^2 + 4kb \geq 0$ $4k^2 - 4bk - a^2 \leq 0$ Find critical values $k = \frac{4b \pm \sqrt{16b^2 - 4(4)(-a^2)}}{2(4)}$ $k = \frac{b \pm \sqrt{b^2 + a^2}}{2}$ Therefore $4k^2 - 4bk - a^2 \leq 0$ $\frac{b - \sqrt{b^2 + a^2}}{2} \leq k \leq \frac{b + \sqrt{b^2 + a^2}}{2}$ Since $-1 \leq k \leq 4$, $\frac{b}{2} = \frac{-1+4}{2} \Rightarrow b = 3$ $\frac{b + \sqrt{b^2 + a^2}}{2} = 4 \Rightarrow \sqrt{9 + a^2} = 5 \Rightarrow a = \pm 4$	$y = \frac{ax+b}{x^2+1}$ is a continuous curve with 2 turning points. The general shape of the graph will be of the form as shown below.  Thus range of values of y where $y = \frac{ax+b}{x^2+1}$ is defined for would be where the horizontal line meets the maximum and minimum point. Note that if $a = 0$, then the general shape of the graph of $y = \frac{b}{x^2+1}$ would be of the form as shown below  which is not possible in this case. Many students recognized the endpoints of the given inequality as roots of the equation whereby discriminant = 0. However, students should still have started their

Qn	Solution	Comments
		working with the reasoning that discriminant ≥ 0 based on the context given in the question. Immediately starting your working with discriminant $= 0$ suggests that there is only one intersection between the curve and each of the lines $y = -1$ and $y = 4$, but most students did not justify why this is the case in the question. Students need to indicate the correct sign of the discriminant to be credited with the method mark. Some students were confused about the restriction on k and incorrectly rejected $a = -4$.
4(i)	Let $a_1 = b_1 = a$ $a_3 = b_3 \Rightarrow a + 2d = a r^2 \quad \text{-----}(1)$ $a_7 = b_5 \Rightarrow a + 6d = a r^4 \quad \text{-----}(2)$ Eliminate d, $(1) \times 3 - (2)$: $2a = 3a r^2 - a r^4$ $a(r^4 - 3r^2 + 2) = 0$ Since $a \neq 0$, $r^4 - 3r^2 + 2 = 0 \quad \text{-----}(3)$ $\Rightarrow r^2 = 1, \text{ or } r^2 = 2$ Since $r > 0$, $r \neq 1$, therefore, $r = \sqrt{2}$. Sub $r = \sqrt{2}$ into (1): $a + 2d = 2a \Rightarrow d = \frac{a}{2}$	Students need to accustom themselves to dealing with multiple unknowns such as $a_1, a_2, a_3, b_1, b_2, b_3$ and approach such questions step by step starting with info given to reduce the number of unknowns. In this case there is a need to apply the formula for nth term of AP and GP to obtain (1) and (2). This leaves the simultaneous equations with three unknowns a, d and r. Thereafter students can choose either to eliminate d for instance to formulate $r^4 - 3r^2 + 2 = 0$ The last part involves using $d = \frac{a}{2}$ and $r = \sqrt{2}$ to express n in terms of m.

Qn	Solution	Comments
	If $a_n = b_m$: $a + (n-1)d = ar^{m-1}$ $a + (n-1)\frac{a}{2} = ar^{m-1}$ $1 + \frac{(n-1)}{2} = r^{m-1}$ $\frac{(n+1)}{2} = 2^{\frac{1}{2}(m-1)}$ $n = 2^{\frac{1}{2}(m+1)} - 1$	
4(ii)	Since $n \in \mathbb{Z}^+$, $2^{\frac{1}{2}(m+1)} - 1 = 1, 3, 7, 15, \dots$ $2^{\frac{1}{2}(m+1)} = 2, 4, 8, 16, \dots$ $\frac{1}{2}(m+1) = 1, 2, 3, 4, \dots$ $m = 1, 3, 5, 7, \dots$ For n to be an integer, m must be a positive odd integer.	It's easier to list out all possible values of n and work backwards to identify the condition for m. Many students left out the word 'odd'.
5(a)	Area of $ABC = \frac{1}{2} \overrightarrow{AC} \times \overrightarrow{BC}$ $= \frac{1}{2} (-\sqrt{2}\mathbf{a} - \mathbf{b} - \mathbf{a}) \times (-\sqrt{2}\mathbf{a} - \mathbf{b} - \mathbf{b}) $ $= \frac{1}{2} ((- \sqrt{2} - 1)\mathbf{a} - \mathbf{b}) \times (-\sqrt{2}\mathbf{a} - 2\mathbf{b}) $ $= \frac{1}{2} (2 + \sqrt{2}) \mathbf{a} \times \mathbf{a} + \sqrt{2} \mathbf{b} \times \mathbf{a} + (2\sqrt{2} + 2) \mathbf{a} \times \mathbf{b} + 2 \mathbf{b} \times \mathbf{b} $ $= \frac{1}{2} \sqrt{2} \mathbf{b} \times \mathbf{a} + (2\sqrt{2} + 2) \mathbf{a} \times \mathbf{b} $ $= \frac{1}{2} (\sqrt{2} + 2) \mathbf{a} \times \mathbf{b} $ $= \frac{\sqrt{2} + 2}{2} \mathbf{a} \times \mathbf{b} $ $\therefore k = \frac{\sqrt{2} + 2}{2}$ Since $\angle AOB = 2 \times \angle ACB$ since $\angle$ at centre $= 2 \times \angle$ at circumference.	Mistakes made include: - Wrong formula for area of triangle where $\frac{1}{2}$ was often missed out. $\mathbf{a} \times \mathbf{a} \neq \mathbf{a} ^2$, $\mathbf{b} \times \mathbf{b} \neq \mathbf{b} ^2$ $\mathbf{b} \times \mathbf{a} \neq \mathbf{a} \times \mathbf{b}$ Students are required to recall geometry learnt in Secondary school and take $\angle AOB = \frac{\pi}{3}$.

Qn	Solution	Comments
	$\begin{aligned}\text{Area of } ABC &= \mathbf{a} \times \mathbf{b} \\ &= \mathbf{a} \mathbf{b} \left \sin \frac{2\pi}{3}\right \hat{n} \\ &= (1)(1)\left(\frac{\sqrt{3}}{2}\right)(1) \\ &= \frac{\sqrt{3}}{2} \text{ units}^2\end{aligned}$	
5(b)	Since $\overrightarrow{MC} = 3\overrightarrow{BC}$, $\overrightarrow{OB} = \frac{\overrightarrow{OM} + 2\overrightarrow{OC}}{3}$ $\overrightarrow{OM} = 3\overrightarrow{OB} - 2\overrightarrow{OC}$ $= 3\mathbf{b} - 2\left(-\frac{2}{3}\mathbf{a} - \frac{1}{3}\mathbf{b}\right)$ $= \frac{4}{3}\mathbf{a} + \frac{11}{3}\mathbf{b}$ Alternatively, $\begin{aligned}\overrightarrow{OM} &= \overrightarrow{OC} + \overrightarrow{CM} \\ &= \overrightarrow{OC} + 3\overrightarrow{CB} \\ &= 3\overrightarrow{OB} - 3\overrightarrow{OC}\end{aligned}$ 	Students should try drawing out the triangle to ensure labelling the correct ratio based on $\overrightarrow{MC} = 3\overrightarrow{BC}$. Alternatively just apply simple vectors additions/subtractions and apply $\overrightarrow{MC} = 3\overrightarrow{BC}$.
6(i)	$\begin{aligned}\frac{d}{d\theta} \sec^n \theta &= n \sec^{n-1} \theta (\sec \theta \tan \theta) \\ &= n \sec^n \theta \tan \theta\end{aligned}$	Use chain rule. Simplify the expression by grouping like terms together. This helps in identifying the usefulness of this result for (iii).
6(ii)	$x = 2 \tan^2 \theta \Rightarrow \frac{dx}{d\theta} = 4 \tan \theta \sec^2 \theta$ When $x = 0$, $\tan \theta = 0 \Rightarrow \theta = 0$ When $x = 2$, $\tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$	Here $\sqrt{\sec^2 \theta} = \sec \theta = \sec \theta$ due to the range of values of θ. This question was forgiving to just take the positive root. Students need to be mindful of this commonly neglected result: $\sqrt{x^2} = x$, as x may be negative, but LHS is always non-negative. Do be reminded to change all the variables, including the limits, to be in terms of θ in a single step. A handful of students got stuck at the step $\int_0^{\frac{\pi}{4}} 8 \tan^3 \theta \sec^3 \theta d\theta$. Students need to be familiar with the

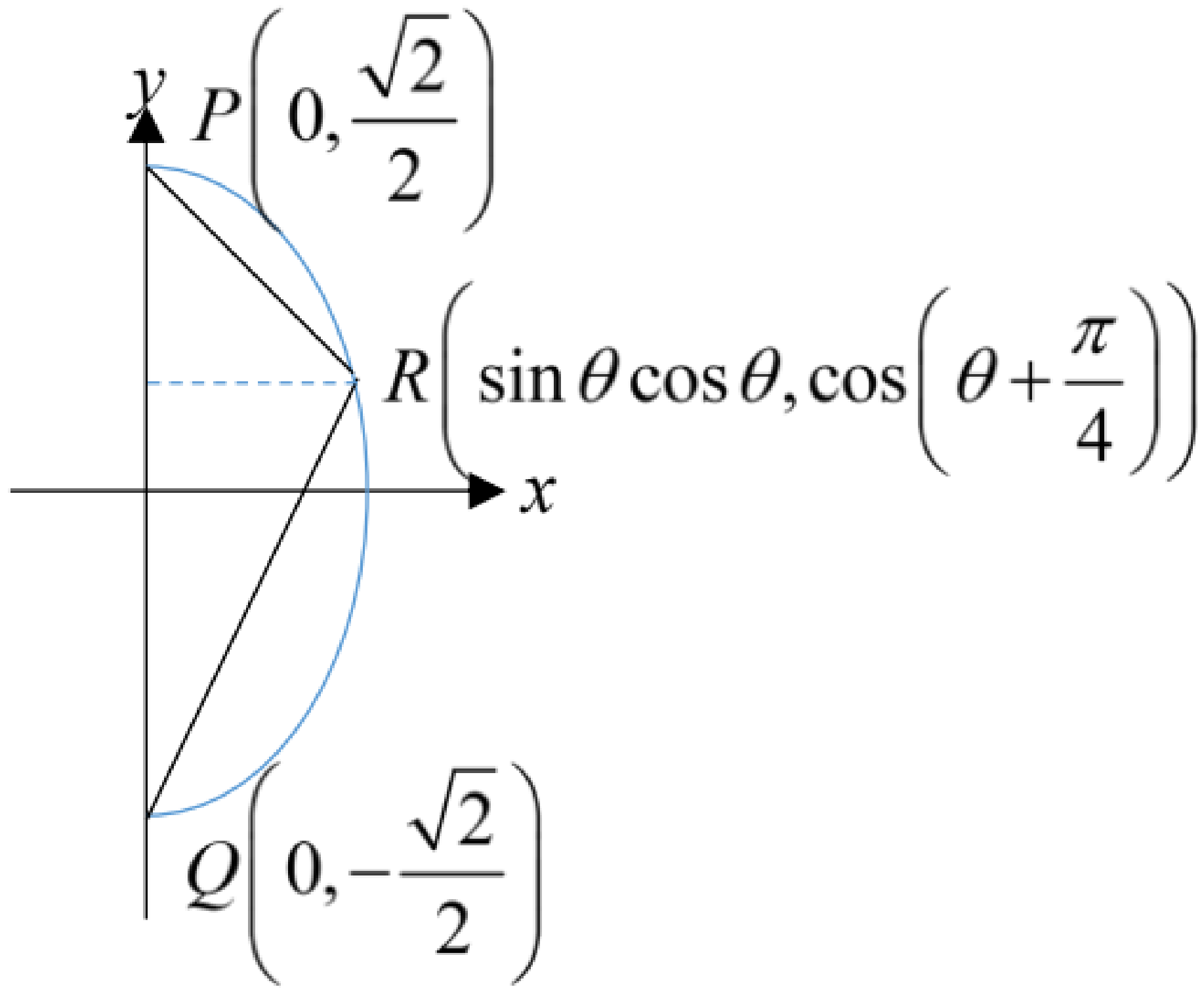
Qn	Solution	Comments
	$\int_0^2 x \sqrt{1 + \frac{x}{2}} \, dx$ $= \int_0^{\frac{\pi}{4}} 2 \tan^2 \theta \sqrt{1 + \tan^2 \theta} (4 \tan \theta \sec^2 \theta) \, d\theta$ $= \int_0^{\frac{\pi}{4}} 8 \tan^3 \theta \sec \theta \sec^2 \theta \, d\theta$ $= \int_0^{\frac{\pi}{4}} 8 \tan^3 \theta \sec^3 \theta \, d\theta \left(\because \sec \theta > 0 \text{ for } 0 < \theta < \frac{\pi}{4} \right)$ $= 8 \int_0^{\frac{\pi}{4}} \tan \theta (\sec^2 \theta - 1) \sec^3 \theta \, d\theta$ $= 8 \int_0^{\frac{\pi}{4}} \sec^5 \theta \tan \theta - \sec^3 \theta \tan \theta \, d\theta$ where $k = 8$, $a = 0$, $b = \frac{\pi}{4}$	use of trigonometric identities that may not be in MF26.
6(iii)	$\int_0^2 x \sqrt{1 + \frac{x}{2}} \, dx$ $= 8 \int_0^{\frac{\pi}{4}} \sec^5 \theta \tan \theta - \sec^3 \theta \tan \theta \, d\theta$ $= 8 \left[\frac{\sec^5 \theta}{5} - \frac{\sec^3 \theta}{3} \right]_0^{\frac{\pi}{4}}$ $= 8 \left(\frac{(\sqrt{2})^5}{5} - \frac{(\sqrt{2})^3}{3} \right) - 8 \left(\frac{1}{5} - \frac{1}{3} \right)$ $= \frac{16}{15} (\sqrt{2} + 1)$ $A = \frac{16}{15}, B = 1$	This question is a follow-on from (i) and (ii). Students may have missed the cue from (i) if they did not simplify their expression earlier. Many students had difficulty simplifying the surds. Take note that $\sqrt{2}^5 = 4\sqrt{2}$. Do be reminded to present the answer to the question's requirement, $A(\sqrt{2} + 1)$. Again, students are required to simplify and group the like terms together.
7(i)	To find exact points of intersection, sub C_2 into C_1 : $x^2 + \frac{4}{1+x^2} = 4$ $x^2 + x^4 + 4 = 4 + 4x^2$ $x^4 - 3x^2 = 0$ $x^2(x^2 - 3) = 0$ $x = 0 \text{ or } x = \pm\sqrt{3}$ (or using G.C.)	Many students omitted the root $x = 0$. Students should carefully look at the diagram to see that there are 6 points of intersection, and check back with their answer.

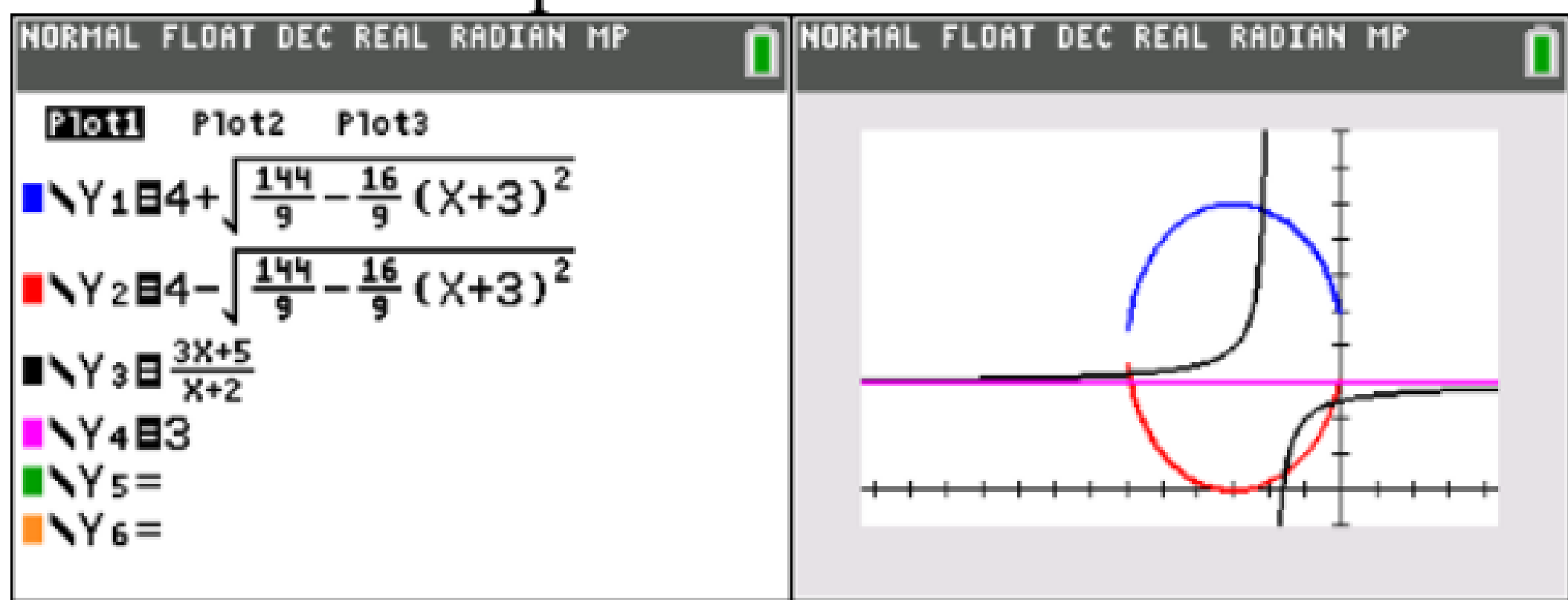
Qn	Solution	Comments
	When $x = 0$, $y^2 = 4 \Rightarrow y = \pm 2$ When $x = \pm\sqrt{3}$, $y^2 = 1 \Rightarrow y = \pm 1$ $\therefore (0, 2), (0, -2), (\sqrt{3}, 1), (-\sqrt{3}, 1), (\sqrt{3}, -1), (-\sqrt{3}, -1)$	
7(ii)	Method 1: Using $\int x \, dy$ Area of shaded region $= 4 \left[\int_0^1 \sqrt{4-y^2} \, dy + \int_1^2 \sqrt{\frac{4}{y^2}-1} \, dy \right]$ $= 11.260 \text{ units}^2 \text{ (to 3d.p)}$ OR $= 4\pi - 4 \int_1^2 \sqrt{4-y^2} - \sqrt{\frac{4}{y^2}-1} \, dy$ Method 2: Using $\int y \, dx$ Area of shaded region $= 4 \left[\int_0^{\sqrt{3}} \frac{2}{\sqrt{1+x^2}} \, dx + \int_{\sqrt{3}}^2 \sqrt{4-x^2} \, dx \right]$ $= 11.260 \text{ units}^2 \text{ (to 3d.p)}$ OR $= 4\pi - 2 \int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{4-x^2} - \sqrt{\frac{4}{1+x^2}} \, dx$	There are many permutations of solving this question. The key note of this question is to first identify the boundaries of the shaded region carefully, and then realize that the shaded region is symmetrical both about the x and the y axes. Therefore, students may find the area in one quadrant and then multiply by 4. Some students took the area of the circle to minus the area of the unshaded region, which also gives the answer. Students need to be mindful of the question requirement: to 3 decimal places. While it gives students the hint to use GC, some students neglected the hint and solved by hand, giving erroneous solutions as the solving is out of syllabus. Some students also forgot to leave their answers in 3 dp, and were penalized at the front page.
7(iii)	$V_x = 2\pi \left[\int_0^{\sqrt{3}} \frac{4}{1+x^2} \, dx + \int_{\sqrt{3}}^2 4-x^2 \, dx \right]$ $= 2\pi \left[\left[4 \tan^{-1} x \right]_0^{\sqrt{3}} + \left[4x - \frac{x^2}{3} \right]_{\sqrt{3}}^2 \right]$ $= 2\pi \left(\frac{4\pi}{3} + 8 - \frac{8}{3} - 4\sqrt{3} + \sqrt{3} \right)$ $= \frac{2\pi}{3} (4\pi + 16 - 9\sqrt{3}) \text{ units}^3$ OR $V_x = \frac{4}{3} \pi (2)^3 - \pi \left[\int_{-\sqrt{3}}^{\sqrt{3}} 4-x^2 - \frac{4}{1+x^2} \, dx \right]$	A handful of students were misled by (ii), where they multiplied the area by 4. This time the area in the 1st quadrant is rotated 2π radian in the x-axis, forming half of the required volume. Therefore, students should only multiply this volume by 2. Many students took the volume of the sphere minus the hollow region, which was also correct. A small number of students integrated with respect to y-axis,

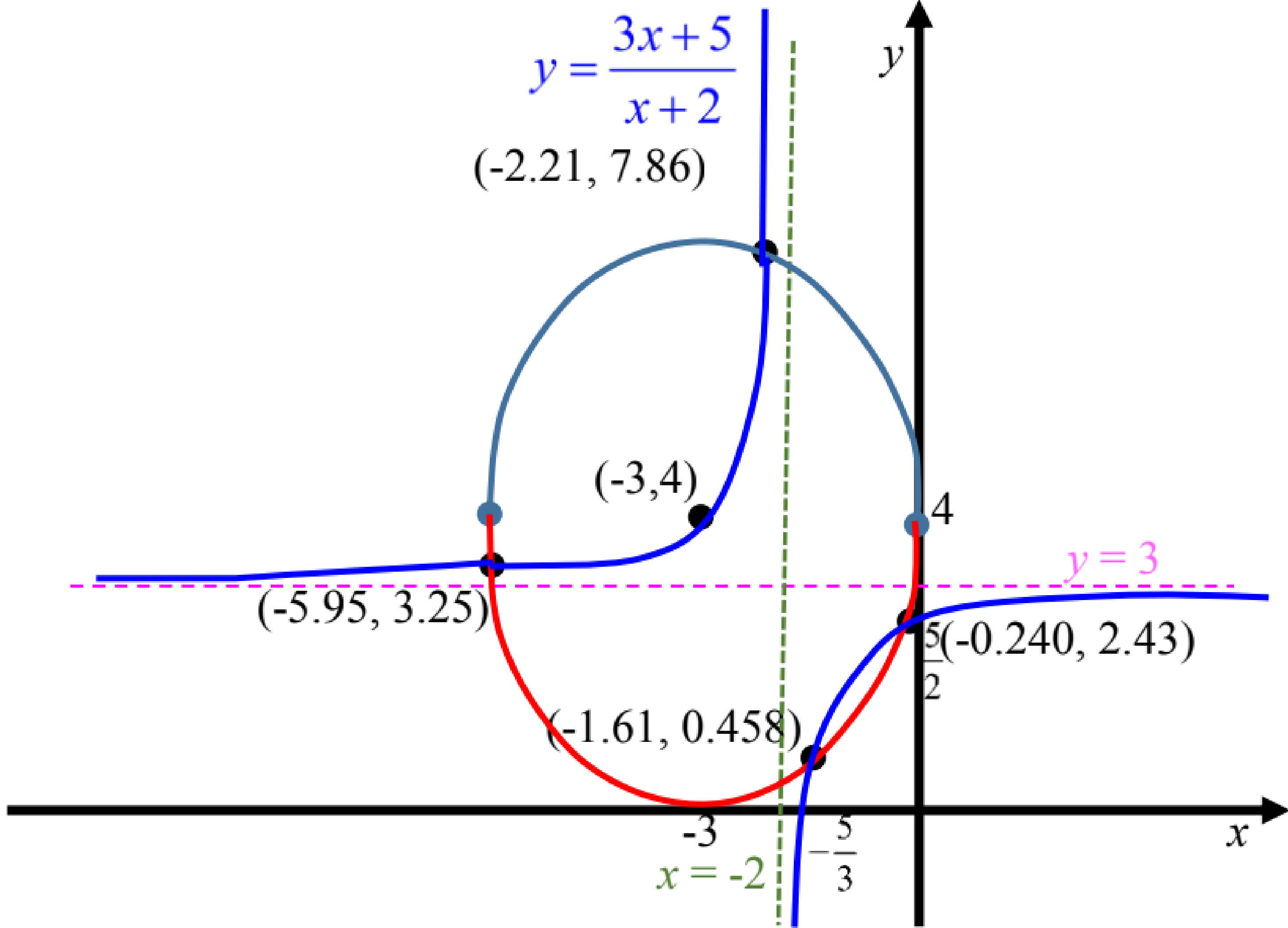
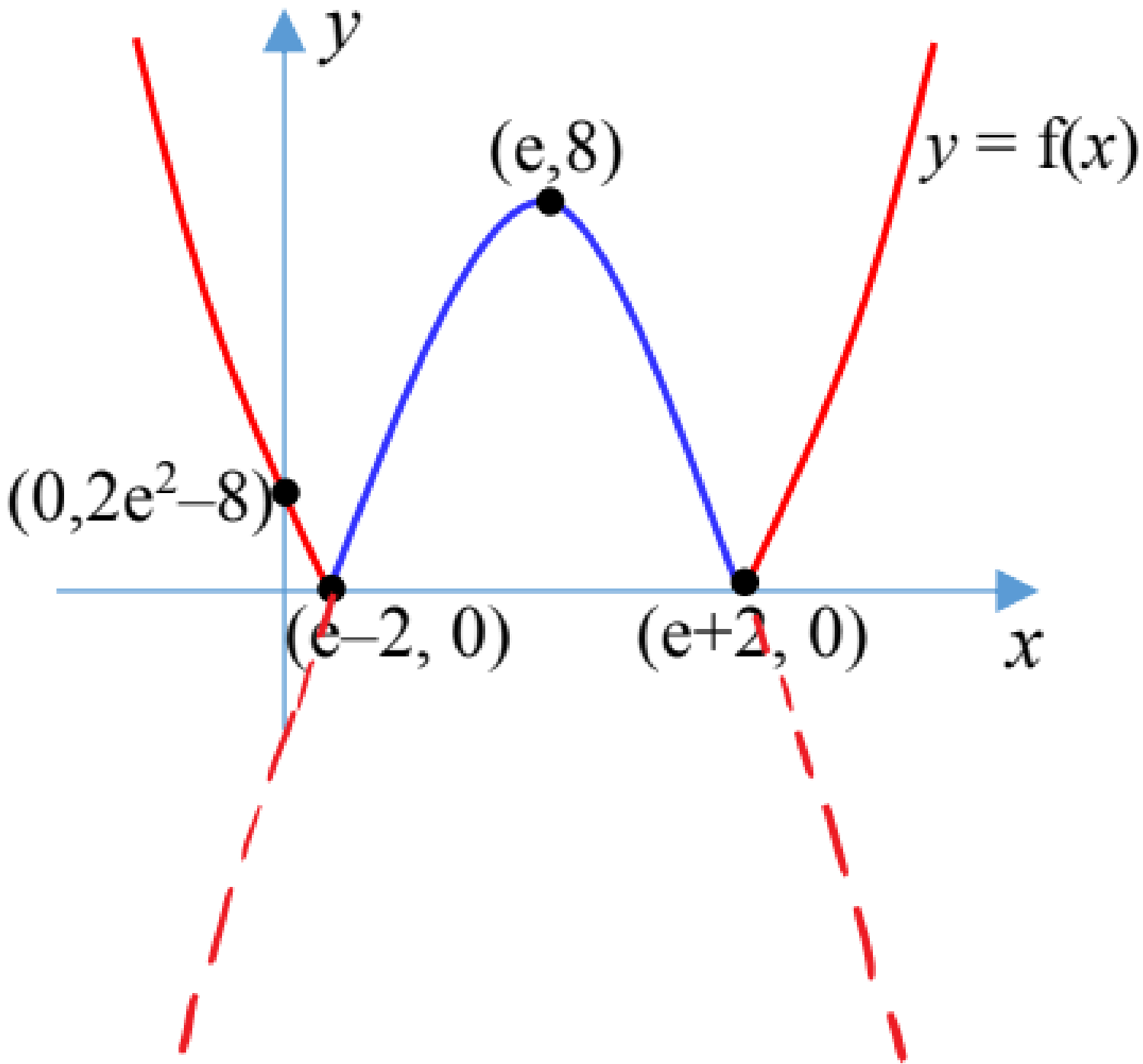
Qn	Solution	Comments
		which does not give the same volume.
8(i)	From RHS: $\frac{1}{2} \left(\frac{1}{\sqrt{2r+1}} - \frac{1}{\sqrt{2r+3}} \right) = \frac{1}{2} \frac{\sqrt{2r+3} - \sqrt{2r+1}}{\sqrt{2r+1}\sqrt{2r+3}}$ $= \frac{1}{2} \frac{(\sqrt{2r+3} - \sqrt{2r+1})(\sqrt{2r+3} + \sqrt{2r+1})}{\sqrt{2r+1}\sqrt{2r+3}(\sqrt{2r+3} + \sqrt{2r+1})}$ $= \frac{1}{2} \frac{(2r+3) - (2r+1)}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$ $= \frac{1}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$ =LHS OR From LHS: $\frac{1}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$ $= \frac{1}{\sqrt{(2r+3)}\sqrt{(2r+1)} [\sqrt{(2r+3)} + \sqrt{(2r+1)}]}$ $= \frac{\sqrt{(2r+3)} - \sqrt{(2r+1)}}{\sqrt{(2r+3)}\sqrt{(2r+1)} [2r+3 - 2r-1]}$ $= \frac{\sqrt{2r+3} - \sqrt{2r+1}}{2\sqrt{2r+1}\sqrt{2r+3}}$ $= \frac{1}{2} \left(\frac{1}{\sqrt{2r+1}} - \frac{1}{\sqrt{2r+3}} \right)$ =RHS	Many students made weak attempts for part (i) or left it blank. This is not a question on partial fractions, rather it involves the idea of multiplication by a conjugate surd. There were also algebraic misconceptions. Please note that : $\frac{2r+3}{(2r+3)\sqrt{(2r+1)}} - \frac{2r+1}{(2r+1)\sqrt{(2r+3)}}$ $\neq \frac{2}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$
8(ii)	$S_n = \sum_{r=1}^n \frac{1}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$ $= \frac{1}{2} \sum_{r=1}^n \left(\frac{1}{\sqrt{(2r+1)}} - \frac{1}{\sqrt{(2r+3)}} \right) \quad \text{-----(1)}$	This part was well done by the majority of students. Students should show cancellation for at least the first two rows and last two rows. A small minority made errors such as writing the final answer erroneously as $\frac{1}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2r+3}} \right)$

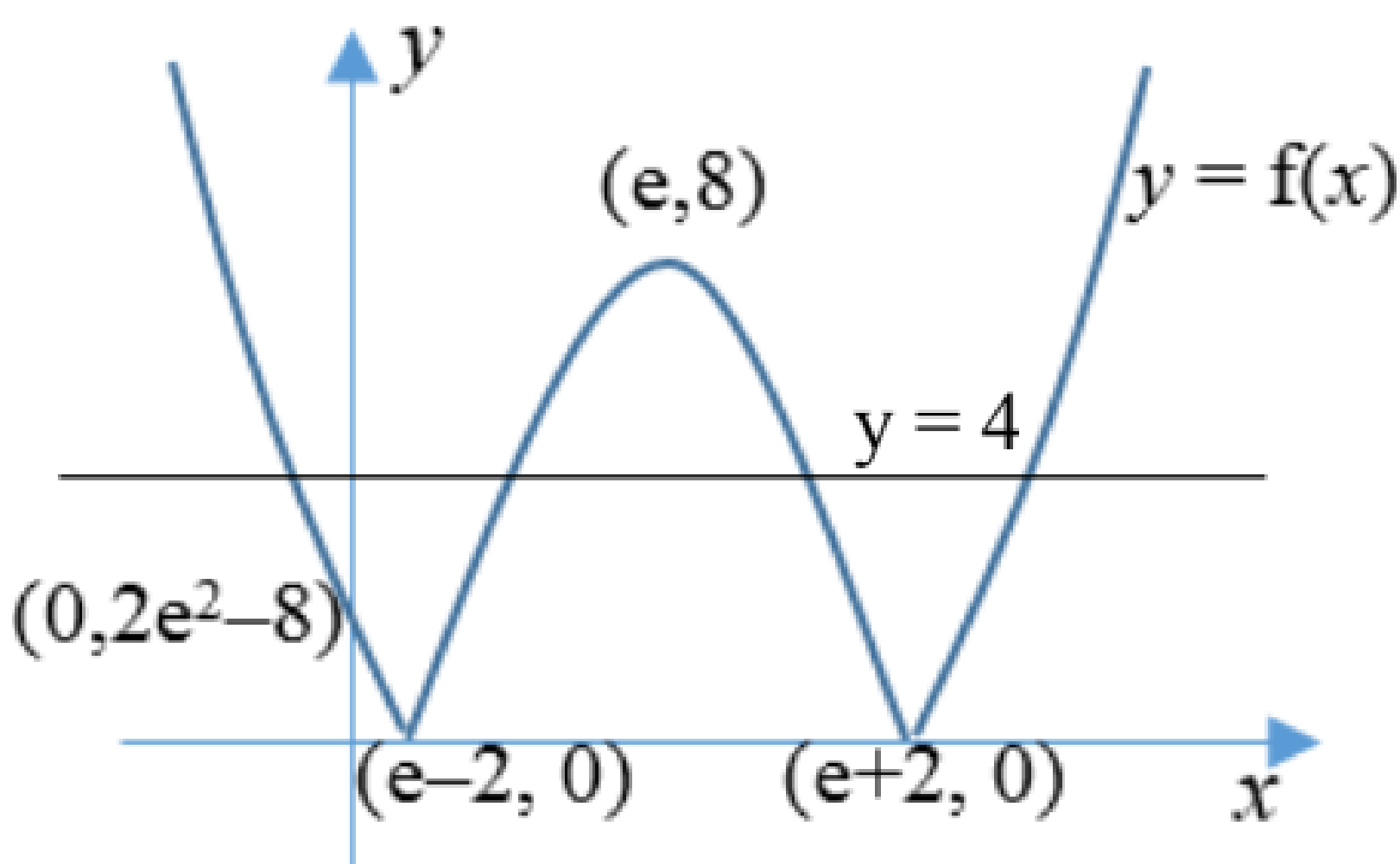
Qn	Solution	Comments
	$= \frac{1}{2} \left[\begin{array}{c} \frac{1}{\sqrt{3}} - \frac{1}{\sqrt{5}} \\ \frac{1}{\sqrt{5}} - \frac{1}{\sqrt{7}} \\ \dots \\ \frac{1}{\sqrt{2n-1}} - \frac{1}{\sqrt{2n+1}} \\ \frac{1}{\sqrt{2n+1}} - \frac{1}{\sqrt{2n+3}} \end{array} \right]$ $= \frac{1}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2n+3}} \right)$	or $\frac{1}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2N+3}} \right)$ instead of $\frac{1}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2n+3}} \right)$.
8(iii)	$n \rightarrow \infty, \frac{1}{\sqrt{2n+3}} \rightarrow 0$ $\lim_{n \rightarrow \infty} S_n = \frac{1}{2\sqrt{3}}$ $\left S_n - \frac{1}{2\sqrt{3}} \right < 0.05$ $\left \frac{1}{2} \left(\frac{1}{\sqrt{3}} - \frac{1}{\sqrt{2n+3}} \right) - \frac{1}{2\sqrt{3}} \right < 0.05 \quad \text{-----}(2)$ $\left \frac{1}{2\sqrt{2n+3}} \right < 0.05$ $\frac{1}{2\sqrt{2n+3}} < 0.05$ $\sqrt{2n+3} > 10$ $2n+3 > 100$ $n > 48.5$ OR From GC, $n = 48, \frac{1}{2\sqrt{2n+3}} = 0.05025$ $n = 49, \frac{1}{2\sqrt{2n+3}} = 0.04975$ therefore the least value n is 49.	Some students misunderstood this series as a geometric series, quoting the formula $S_\infty = \frac{a}{1-r}$ which should not be used here. The question should translate to $S_\infty - S_n < 0.05$ or $S_\infty - S_n < 0.05$ as $S_\infty > S_n$ here. Some of those who obtained the inequality went through laborious methods to solve it. The working for solving the inequality should be clearly shown. If the table method is employed, then relevant values computed for $n = 48$ and $n = 49$ should be presented. Values obtained from GC such as 2.5E-4 should be written as 2.5×10^{-4}.
9(i)	$\frac{dx}{dt} = \cos^2 t - \sin^2 t$	This part was generally well done. Note that

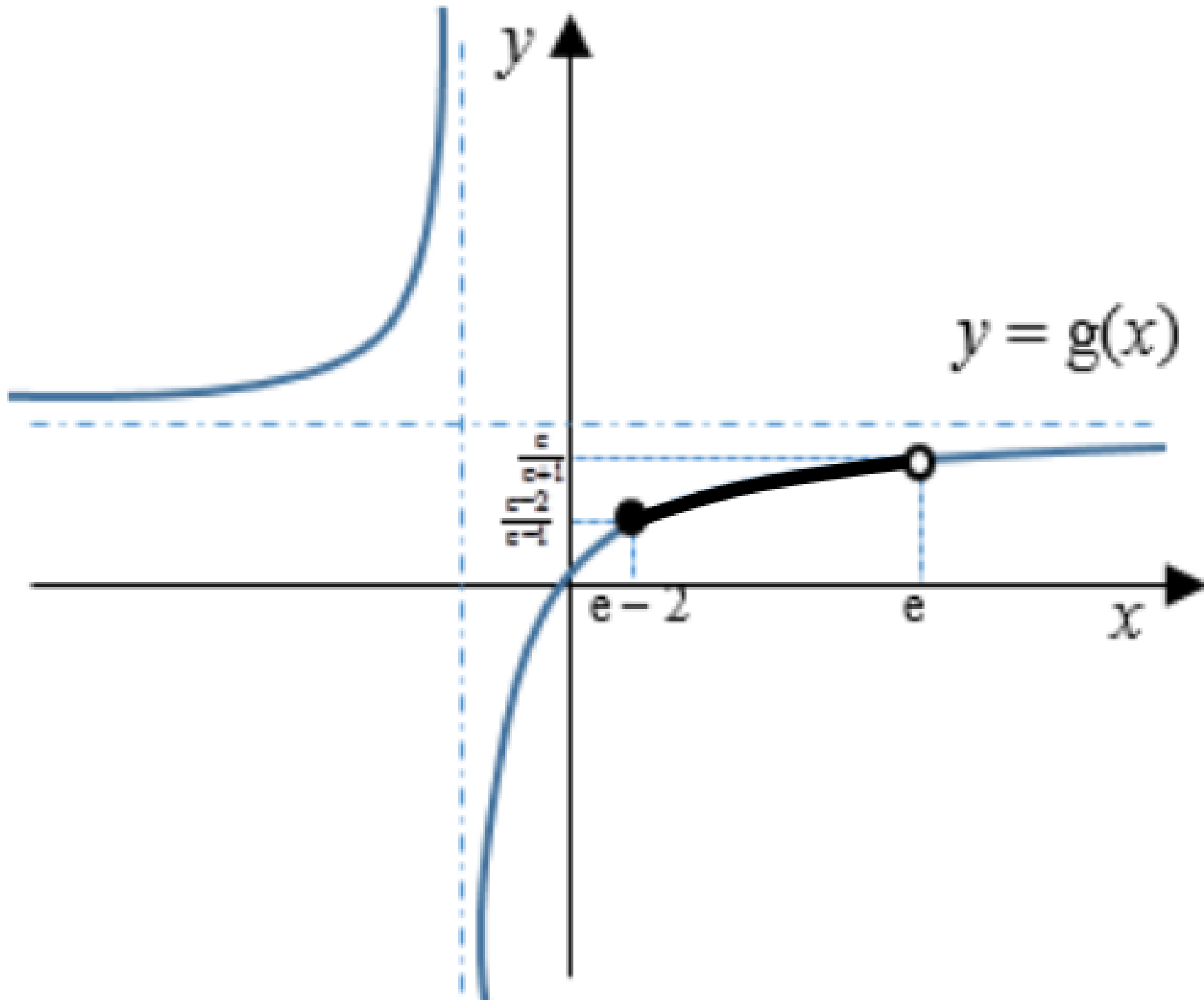
Qn	Solution	Comments
	$\frac{dy}{dt} = \frac{d}{dt} \left(\cos t \cos \frac{\pi}{4} - \sin t \sin \frac{\pi}{4} \right)$ $= \frac{\sqrt{2}}{2} \frac{d}{dt} (\cos t - \sin t)$ $= -\frac{\sqrt{2}}{2} (\sin t + \cos t)$ $\therefore \frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx}$ $= \frac{-\frac{\sqrt{2}}{2} (\sin t + \cos t)}{\cos^2 t - \sin^2 t}$ $= -\frac{\sin t + \cos t}{\sqrt{2} (\cos t + \sin t)(\cos t - \sin t)}$ $= \frac{1}{\sqrt{2} (\sin t - \cos t)} \text{ (shown)}$	$\sin \left(t + \frac{\pi}{4} \right) \neq \sin t + \sin \left(\frac{\pi}{4} \right)$ Students should rewrite the expression using the addition formula in MF26 where $\sin(A + B) = \sin A \cos B + \cos A \sin B$
9(ii)	For tangent parallel to y – axis, $\frac{dy}{dx} \text{ is undefined} \Rightarrow \sin t - \cos t = 0$ $\tan t = 1$ $t = \frac{\pi}{4}$ $\Rightarrow$ Equation of tangent is: $x = \sin \frac{\pi}{4} \cos \frac{\pi}{4} = \left(\frac{\sqrt{2}}{2} \right)^2 = \frac{1}{2}$ $\therefore x = \frac{1}{2}$	This is a ‘hence’ question, so students are required to use the answer from part (i) to solve. When the tangent is parallel to the y-axis, the equation of the tangent is in the form $x = k$, where k is a real number. If the tangent is parallel to the x-axis, the equation of the tangent is in the form $y = h$ where h is a real number.
9(iii)	<u>Method 1</u> When $x = 0$, $\sin t \cos t = 0$ $2 \sin t \cos t = 0$ $\sin 2t = 0$ $2t = 0 \text{ or } \pi$ $t = 0 \Rightarrow y = \cos \left(0 + \frac{\pi}{4} \right) = \frac{\sqrt{2}}{2}$ $\text{or } t = \frac{\pi}{2} \Rightarrow y = \cos \left(\frac{\pi}{2} + \frac{\pi}{4} \right) = -\frac{\sqrt{2}}{2}$	The equation $\sin t \cos t = 0$ gives two t values one for the coordinates of P and one for the coordinates of Q. Note that $\cos \left(\frac{\pi}{4} \right)$ is only equal to $\frac{1}{\sqrt{2}}$ but not $-\frac{1}{\sqrt{2}}$. $-\frac{1}{\sqrt{2}}$ is obtained when $t = \frac{\pi}{2}$

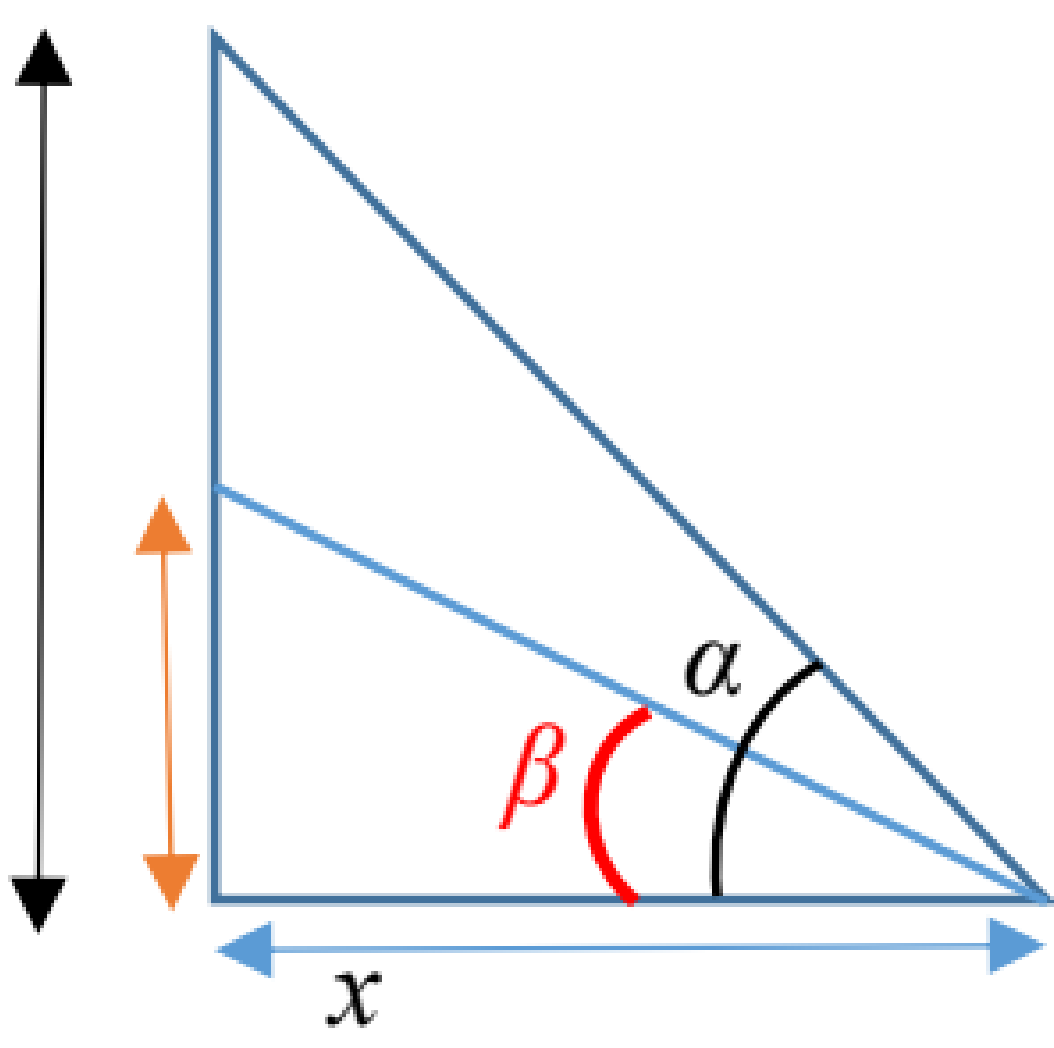
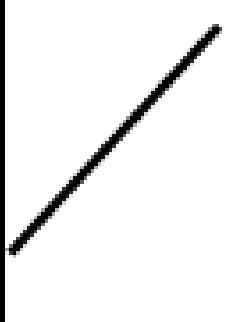
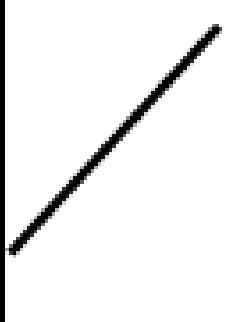
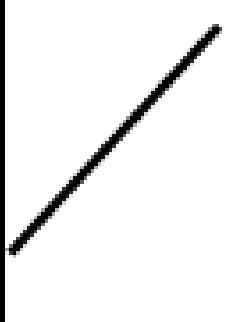
Qn	Solution	Comments
	<u>Method 2</u> When $x = 0$, $\sin t \cos t = 0$ $\sin t = 0 \Rightarrow t = 0 \Rightarrow y = \cos\left(0 + \frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$ $\cos t = 0 \Rightarrow t = \frac{\pi}{2} \Rightarrow y = \cos\left(\frac{\pi}{2} + \frac{\pi}{4}\right) = -\frac{\sqrt{2}}{2}$ $\therefore P\left(0, \frac{\sqrt{2}}{2}\right), Q\left(0, -\frac{\sqrt{2}}{2}\right)$	
9(iv)	 From the diagram, area of triangle PQR is given by $\begin{aligned} & \frac{1}{2} \times 2 \left(\frac{\sqrt{2}}{2} \right) \times \sin \theta \cos \theta \\ &= \frac{\sqrt{2}}{2} \left(\frac{1}{2} \sin 2\theta \right) \\ &= \frac{\sin 2\theta}{2\sqrt{2}} \\ &= \frac{\sqrt{2} \sin 2\theta}{4} \text{ (shown)} \end{aligned}$ Since $0 \leq \sin 2\theta \leq 1$ for $0 \leq \theta \leq \frac{\pi}{2}$, max area of triangle PQR is when $\sin 2\theta = 1 \Rightarrow \theta = \frac{\pi}{4}$	This part of the question is generally quite well done. Some of the students used shoelace method but it is not necessary. The method of using differentiation is not recommended, since students are expected to be familiar with trigonometric properties.

Qn	Solution	Comments
10 (i)	$y = \frac{3x+5}{x+2} = 3 - \frac{1}{x+2}$ <u>Method 1:</u> $\frac{1}{x} \xrightarrow{1} \frac{1}{x+2} \xrightarrow{2} -\frac{1}{x+2} \xrightarrow{3} 3 - \frac{1}{x+2}$ 1. Translate 2 units in the negative x-direction 2. Reflect in x-axis 3. Translate 3 units in the positive y-direction <u>Method 2:</u> $\frac{1}{x} \rightarrow -\frac{1}{x} \rightarrow -\frac{1}{x+2} \rightarrow 3 - \frac{1}{x+2}$ 1. Reflect in y-axis 2. Translate 2 units in the negative x-direction 3. Translate 3 units in the positive y-direction Students should recognize the equation of the ellipse. $16(x+3)^2 + 9(y-4)^2 = 144$ $\frac{(x+3)^2}{3^2} + \frac{(y-4)^2}{4^2} = 1$ When using GC, student should use zoon-square to ensure that the axis are both equal in scale. 	Some students did not express $y = \frac{3x+5}{x+2} = 3 - \frac{1}{x+2}$, hence it would be difficult for them to describe the sequence of transformations. Students should use ‘translate’ or ‘reflect’ to describe the transformations and not use words like ‘shift’, ‘move’ or ‘flip’. Students were generally confused with the transformation of $y = f(-x)$ which is the reflection of $f(x)$ in the y-axis. Likewise, the transformation of $y = -f(x)$ is the reflection of $y = f(x)$ in the x-axis. The graph of $16(x+3)^2 + 9(y-4)^2 = 144$ is an ellipse. It is symmetrical about the line $y = 4$, $x = -3$ and touches the x-axis and y-axis. It is good to note that the curve $y = 3 - \frac{1}{x+2}$ passes through $(-3, 4)$ which is also the centre of the ellipse. Students should sketch the ellipse with its vertical radius longer than its horizontal radius. Also, some students sketched a circle which was not acceptable.

Qn	Solution	Comments
10 (ii)	 From the graph, compare the curve with the lower half of the ellipse, $\{x \in \mathbb{R} : -5.95 < x < -2 \text{ or } -1.61 < x < -0.240\}$	For curve sketching, there are - 2 asymptotes $y = 3$, $x = -2$. - 4 axial intercepts $(0, 4)$, $(0, \frac{5}{2})$, $(-\frac{5}{3}, 0)$, $(-3, 0)$ - 4 points of intersection of the two graphs Not many students answered this correctly. Also, the question stated that the answer must be in set notation.
11 (i)	 To find exact x-coordinates of x intercepts: $4 - (x - e)^2 = 0$ $(x - e)^2 = 4$ $x - e = \pm 2$ $x = e \pm 2$ To find exact y-coordinates of y intercept: $2[-4 + (0 - e)^2] = y$ $y = 2e^2 - 8$	When sketching the modulus function, students need to note that the question requires coordinates in exact form. $y = 2 4 - (0 - e)^2 = 2 4 - e^2 $ Note that $4 - e^2 < 0$ and therefore $2e^2 - 8 < 8$ Do note that the portion of the quadratic function of $y = 2[4 - (x - e)^2]$ below the x-axis would be reflected due to the modulus sign and hence the two “reflected portions” in red should not be drawn as straight lines.

Qn	Solution	Comments
	NOTE: $2 4 - (x - e)^2 = \begin{cases} 2[4 - (x - e)^2] & , e - 2 \leq x \leq e + 2 \\ 2[-4 + (x - e)^2] & , x < e - 2 \text{ or } x > e + 2 \end{cases}$	
11 (ii)	Since $f(e - 2) = f(e + 2)$ Therefore f is not a one to one function. Hence f^{-1} does not exist.  Alternatively, Since the horizontal line $y = 4$ (give a specific counter-example) intersects $y = f(x)$ more than once, therefore f is not a one to one function. Hence f^{-1} does not exist.	To show inverse does not exist, providing a single counter-example is sufficient. However, students should specify clearly the equation of the horizontal line and insert it on the sketch in part (i). Do not write statements such as “the horizontal line $y = k$ passes through $y = f(x)$ at most once” without stating the value of k. Do not write statement such as “the horizontal line $y = 4$ passes through $y = f(x)$ two times” when it should be actually four in this case.
11 (iii)	The function f has an inverse if its domain is restricted to $a \leq x < e$. Smallest $a = e - 2$ Therefore $y = -2(x - e)^2 + 8$ $-\frac{y}{2} = (x - e)^2 - 4$ $(x - e)^2 = 4 - \frac{y}{2}$ $x - e = \pm \sqrt{4 - \frac{y}{2}}$ $x = e \pm \sqrt{4 - \frac{y}{2}}$ Since $e - 2 \leq x < e$ $x = e - \sqrt{4 - \frac{y}{2}}$ $f^{-1} : x \mapsto e - \sqrt{4 - \frac{x}{2}}, \quad 0 \leq x < 8$	Note that exact value of a is required. This answer could be obtained directly from the sketch if students had already left the answer of the x-intercepts in exact form. The function f has an inverse if its domain is restricted to $e - 2 \leq x < e$. Therefore we only need to consider $y = -2(x - e)^2 + 8$ Thus there is no need to consider $y = 2(x - e)^2 - 8$ Students are required to state the reason for rejecting the positive square root, $x = e + \sqrt{4 - \frac{y}{2}}$ due to the restricted domain $e - 2 \leq x < e$

Qn	Solution	Comments
		Question requires similar form of the inverse function, and hence domain of f^{-1} must be stated.
11 (iv)	From part (iii), $e - 2 \leq x < e$ $R_{f^{-1}} = D_f = [e - 2, e)$ $D_g = \mathbb{R} \setminus \{-1\}$ Since $R_{f^{-1}} \subseteq D_g$, gf^{-1} exists  $R_{gf^{-1}} = \left[\frac{e-2}{e-1}, \frac{e}{e+1} \right)$	Students should know that $R_{f^{-1}} = D_f = [e - 2, e)$ since the domain of f is restricted in part (iii). Also, many students were not careful with the inequality and wrote $R_{f^{-1}} = D_f = [e - 2, e]$, wrongly including e in the range of f^{-1} Many students were not able to write $D_g = \mathbb{R} \setminus \{-1\}$ in set notation correctly. To find the $R_{gf^{-1}}$, many simply substituted the end points into the rule of g to get the correct answer without sketching the graph. This is not always correct. It worked in this case because the graph of g is strictly increasing. Students should either sketch the graph or check that the function is strictly increasing.
12 (i)	Let the angle α and β be such that $\tan \alpha = \frac{\frac{y}{2} - 4.5 + 1.5}{x} = \frac{y - 6}{2x}$ and $\tan \beta = \frac{\frac{y}{2} - 4.5 - 1.5}{x} = \frac{y - 12}{2x}$	Some students did not recognize $\tan \theta = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$ which is provided in MF26. Many used $\tan \alpha = \frac{\tan \theta + \tan \beta}{1 - \tan \theta \tan \beta}$ which can be used to prove but it's a much longer method. This is "Show" question, so working needs to be detailed, showing an understanding of the calculations involved.

Qn	Solution	Comments												
	$\tan \theta = \tan (\alpha - \beta)$$= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$= \frac{\frac{y-6}{2x} - \frac{y-12}{2x}}{1 + \left(\frac{y-6}{2x}\right)\left(\frac{y-12}{2x}\right)}$$= \frac{12x}{4x^2 + (y-6)(y-12)}$$= \frac{12x}{4x^2 + A}$ 													
12 (ii)	Differentiate θ implicitly throughout with respect to x $\sec^2 \theta \left(\frac{d\theta}{dx} \right) = \frac{12(4x^2 + A) - 12x(8x)}{[4x^2 + A]^2}$$\sec^2 \theta \left(\frac{d\theta}{dx} \right) = \frac{12[A - 4x^2]}{[4x^2 + A]^2}$$\sec^2 \theta \left(\frac{d\theta}{dx} \right) = \frac{12(\sqrt{A} - 2x)(\sqrt{A} + 2x)}{[4x^2 + A]^2}$ For $\frac{d\theta}{dx} = 0$, $x = \frac{\sqrt{A}}{2} \quad (x > 0)$ Using 1st Derivative Test below, <table><tr><td>x</td><td>$\frac{\sqrt{A}}{2}^-$</td><td>$\frac{\sqrt{A}}{2}$</td><td>$\frac{\sqrt{A}}{2}^+$</td></tr><tr><td>$\frac{d\theta}{dx}$</td><td>+ve</td><td>0</td><td>-ve</td></tr><tr><td>Shape of tangent</td><td></td><td></td><td></td></tr></table> Thus for maximum θ , the player should be $\frac{\sqrt{A}}{2}$ m away from the width of the pool	x	$\frac{\sqrt{A}}{2}^-$	$\frac{\sqrt{A}}{2}$	$\frac{\sqrt{A}}{2}^+$	$\frac{d\theta}{dx}$	+ve	0	-ve	Shape of tangent				Many students missed out “$\sec^2 \theta$” on the LHS when applying implicit differentiation. A good proportion of students were unable to perform quotient rule correctly. Some students did not reject $-\frac{\sqrt{A}}{2}$ or did not use the correct reasoning to reject. Presentation for 1st Derivative Test is generally sloppy. Some students substituted in specific values for x which is incorrect. Students should attempt to factorise the expression of $[A - 4x^2]$ in the numerator to $(\sqrt{A} - 2x)(\sqrt{A} + 2x)$ which allows them to identify the sign of $\frac{d\theta}{dx}$ for 1st Derivative Test.
x	$\frac{\sqrt{A}}{2}^-$	$\frac{\sqrt{A}}{2}$	$\frac{\sqrt{A}}{2}^+$											
$\frac{d\theta}{dx}$	+ve	0	-ve											
Shape of tangent														
12 (iii)	$y = 20$ or $A = 112$	Generally done ok but the result will depend on previous part.												

Qn	Solution	Comments
	maximum θ , the player should be $\frac{\sqrt{112}}{2} = 2\sqrt{7}$ m away from the width of the pool.	As it is a “show” question, students should provide more working, showing clearly the substitution they used.
12 (iv)	Using Chain Rule Since $x = 15$, $\tan \theta = \frac{45}{253}$, $A = 112$ and $\frac{dx}{dt} = -50$, $\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt}$ $= \frac{12(112) - 48(15)^2}{[4(15)^2 + 112]^2 \left[1 + \left(\frac{45}{253}\right)^2\right]} \times -50$ $= 0.447 \text{ rad/min (to 3 sf)}$	Many students missed the negative sign in $\frac{dx}{dt} = -50$. As the player is swimming towards the goal, x is decreasing. A good habit for students will be to write down the chain rule they are using, as well as all the values they are substituting in. Method marks can be awarded even if they make a mistake subsequently. However, marks will not be awarded if only the chain rule is stated without any substitution. Most students did not give the correct units. Calculus calculation is usually done in radians.
13 (i)	Since L lies on the lines of AA' and BB', we need to find the equation of these 2 lines. $l_{AA'}: \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -3-2 \\ 13-3 \\ 8-3 \end{pmatrix} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 10 \\ 5 \end{pmatrix}$ $l_{BB'}: \mathbf{r} = \begin{pmatrix} 4 \\ 3 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 9-4 \\ 13-3 \\ 14-4 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 10 \\ 10 \end{pmatrix}$ $\begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 4 \\ 3 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ 10 \\ 10 \end{pmatrix}$ By G.C., $\lambda = -\frac{1}{5}$ and $\mu = -\frac{1}{5}$ $\overrightarrow{OL} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix} - \frac{1}{5} \begin{pmatrix} -5 \\ 10 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$	Students need to remember to use different parameters for different lines. Students must show and not verify. For this question, can use similar triangles to show the position vector of L. $\text{i.e. } \overrightarrow{A'B'} = 6\overrightarrow{AB} \Rightarrow \overrightarrow{LA'} = 6\overrightarrow{LA}$

Qn	Solution	Comments
13 (ii)	$p_1: \mathbf{r} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = k$ Since $\begin{pmatrix} -3 \\ 13 \\ 8 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 13, k = 13.$ $l_{LC}: \mathbf{r} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2-3 \\ 2-1 \\ 5-2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix}$ $\left(\begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right) \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 13$ $1 + \lambda = 13$ $\lambda = 12$ $\overrightarrow{OC'} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} + 12 \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} = \begin{pmatrix} -9 \\ 13 \\ 38 \end{pmatrix}$	An alternative method for this question is to use $\overrightarrow{LC'} = k\overrightarrow{LC}$ and many students notice that the y-component of the position vector of C must be 13.
13 (iii)	Since $(0, 1053, 0)$ is on p_2, shortest distance is $\left \begin{pmatrix} 0-3 \\ 1053-1 \\ 0-2 \end{pmatrix} \cdot \frac{1}{\sqrt{17}} \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix} \right = \frac{1044}{\sqrt{17}} \approx 253 \text{ cm (3 s.f.)}$	Many students forgot to put the modulus sign when finding length. The dot must be clearly shown.
13 (iv)	Solving $p_1: y = 13$ and $p_2: y + 4z = 1053$, $x = x$ $y = 13$ $z = 260$ So, $l: \mathbf{r} = \begin{pmatrix} 0 \\ 13 \\ 260 \end{pmatrix} + s \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, where $s \in \mathbb{R}$.	